

Brainstorming & Idea Prioritization

1. Idea Generation & Brainstorming

We brainstormed meteorology features to determine the best predictors of localized floods. The features considered were Cloud Cover, Annual Rainfall, and seasonal rainfall distributions.

- Cloud Cover (%): Indicates storm cloud visibility.
- June-September (monsoon) Rainfall: The key hydrological factor for watershed overflow.

2. Idea Prioritization Matrix

- High Impact / Low Effort (Priority 1): Pure Python compiled decision rules on June-September precipitation. Delivers 96.55% accuracy, runs instantly without pandas/sklearn on Vercel.
- High Impact / High Effort (Priority 2): Complex XGBoost models requiring 750MB package wheels.
- Low Impact (Priority 3): Using raw winter rainfall data (Jan-Feb).